

END TERM EXAMINATION**FIFTH SEMESTER [B. TECH] JANUARY 2024****Paper Code: CIC-303****Subject: Compiler Design****Time: 3 Hours****Maximum Marks: 75****Note: Attempt five questions including Q. No. 1 which is compulsory.****Select one question from each unit. Assume missing data.**

Q1. Answer the following questions: (5*3=15)

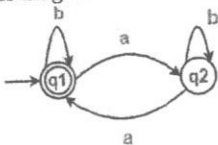
- a) What is Translator? Differentiate between an interpreter and a compiler in terms of functionality.
- b) Define the term "parser" and its role in the context of compiler design.
- c) What is left recursion? Eliminate left recursion from the following grammar:

$$E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow (E) \mid id$$

- d) What is DAG? What are the advantages of DAG?
- e) Consider the transition diagram



Convert the above finite automata into the regular expression.

UNIT-I

- Q2. a) Define the term "compiler" and explain its role in the context of programming languages. Also explain structure and phases of a compiler in detail. (8)
- b) How do we implement lexical analyzer? Explain step by step procedure for implementing lexical analyzer for the following LEX program: (7)

AUXILIARY DEFINITIONS:

(none)

TRANSLATION RULES:

a {} /* Actions are omitted here*/

abb {}

a*b+ {}

- Q3. a) Write step by step procedure to construct the NFA for the following regular expression: (8)

 $R = (a \mid b)^*abb$

- b) What is input buffering? How it helps in construction of compiler? Explain. (7)

P.T.O.**[-2-]****UNIT-II**

- Q4. a) Consider the following grammar: (5)
- $$S \rightarrow ABC, A \rightarrow Aa \mid d, B \rightarrow Bb \mid e, C \rightarrow Cc \mid f$$
- Eliminate left recursion from the above grammar.

- b) Explain and remove the ambiguity from following CFG. (5)
- $$E \rightarrow E+E \mid E-E \mid E/E \mid E^*E \mid (E) \mid -E \mid id$$

- c) Find the FIRST and FOLLOW for the following given grammar: (5)

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' \mid \epsilon$$

$$F \rightarrow (E) \mid id$$

- Q5. a) Test whether the grammar is LL (1) or not and construct a predictive parsing table for it. (8)

$$S \rightarrow iCtSS' \mid a$$

$$S' \rightarrow eS \mid \epsilon$$

$$C \rightarrow b$$

- b) Construct the LR(0) parsing table for the following grammar. (7)

$$S \rightarrow L=R$$

$$S \rightarrow R$$

$$L \rightarrow *R$$

$$L \rightarrow id$$

$$R \rightarrow L$$

Check whether the above grammar is LR (0) grammar or not.

UNIT-III

- Q6. a) State and explain the syntax directed translation scheme for the desk calculator and give the parse tree and translation for the string (8)

 $w = 20 * 7 + 3.$

- b) What is intermediate code representation? Explain various intermediate code representation schemes with the help of an example. (7)

- Q7. a) What do you mean by three address code? Convert the following statements into the Quadruple, Triple and Indirect triple representation: $(A+B)*(C-D^*E)$ (8)
- b) What is type checking? Explain rules of type checking in context of semantic analysis. (7)

UNIT-IV

- Q8. a) What is the use of symbol table? Explain the any two data structures associated with symbol table in detail. (8)
- b) Write short note on the following: (7)
- Loop unrolling & Loop jamming
 - Code generation

- Q9. a) Write short note on the following: (8)
- Basic blocks & flow graph
 - Peephole optimization
- b) Explain the various types of errors generated during the various phases of the compiler. How do we recover from these errors? (7)
